

Geometry
Unit 9
Lesson 8 Practice

Name_____ **Answer Key** ____
Date_____
Period_____

For questions 1-10, round all answers to the nearest thousandth if necessary.

1. Find the lateral area of a pyramid whose height is 8.9 meters, slant height is 6.4 meters, and whose base is a square with a width of 7.6 meters.

$$LA = \frac{1}{2}Ph_s = \frac{1}{2}(30.4)(6.4) = 97.28 \text{ m}^2$$



2. Find the surface area of a pyramid with a height of 8.9 meters, slant height of 6.4 meters, and a square base with a width of 7.6 meters.

$$SA = \frac{1}{2}Ph_s + s^2 = \frac{1}{2}(30.4)(6.4) + (7.6)^2 = 97.28 + 57.76 = 155.04 \text{ m}^2$$

3. Find the lateral area of a rectangular prism with a width of 18.3 feet, a length of 17.5 feet, and a height of 12.4 feet.

$$w = 18.3$$

$$l = 17.5$$

$$h = 12.4$$

$$LA = Ph = (71.6)(12.4) = 887.84 \text{ ft}^2$$



4. Find the surface area of a rectangular prism with a width of 18.3 feet, a length of 17.5 feet, and a height of 12.4 feet.

$$SA = 2B + Ph = 2(18.3)(17.5) + (71.6)(12.4) = 640.5 + 887.84 = 1528.34 \text{ ft}^2$$

5. Find the surface area of a sphere with a diameter of 28.6 centimeters.

$$r = 14.3$$

$$SA = 4\pi r^2 = 4\pi(14.3)^2 \approx 2569.697 \text{ cm}^2 \text{ (when using the pi button)}$$



6. Find the surface area of a hemisphere with a diameter of 92.5 millimeters.

$$r = 46.25$$

$$SA = 3\pi r^2 = 3\pi(46.25)^2 = 20,160.189 \text{ mm}^2$$

7. The surface area of a sphere is 196π inches. What is the diameter of the sphere?

$$SA = 4\pi r^2$$

$$196\pi = 4\pi r^2$$

$$49 = r^2$$

$$r = 7 \text{ in}$$

$$\text{Diameter} = 14 \text{ inches}$$

8. Find the surface area of a cone with a radius of 17.4 yards, a height of 25.3 yards, and a slant height of 30.7 yards.

$$SA = \pi r^2 + \pi r h_s = \pi(17.4)^2 + \pi(17.4)(30.7)$$

$$= 302.76\pi + 534.18\pi \approx 2629.325 \text{ yd}^2 \text{ (when using the pi button)}$$



9. Find the lateral area of a cylinder with a radius of 9.6 inches and a height of 8 inches.

$$LA = 2\pi r h = 2\pi(9.6)(8) \approx 482.549 \text{ in}^2 \text{ (when using the pi button)}$$



10. Find the surface area of a cylinder with a radius of 9.6 inches and a height of 8 inches.

$$SA = 2\pi r^2 + 2\pi r h = 2\pi(9.6)^2 + 2\pi(9.6)(8)$$

$$= 184.32\pi + 153.6\pi = 1061.607 \text{ in}^2 \text{ (when using the pi button)}$$